

Complete MicroPython Installation Guide

For **Windows** and **Linux** Systems

This guide will walk you through setting up MicroPython with Jupyter Notebook on both Windows and Linux operating systems. We'll cover everything from Python installation to ESP32 firmware configuration.

1. Installation Guide for MicroPython Kernel and Jupyter Notebook (**Windows OS**)

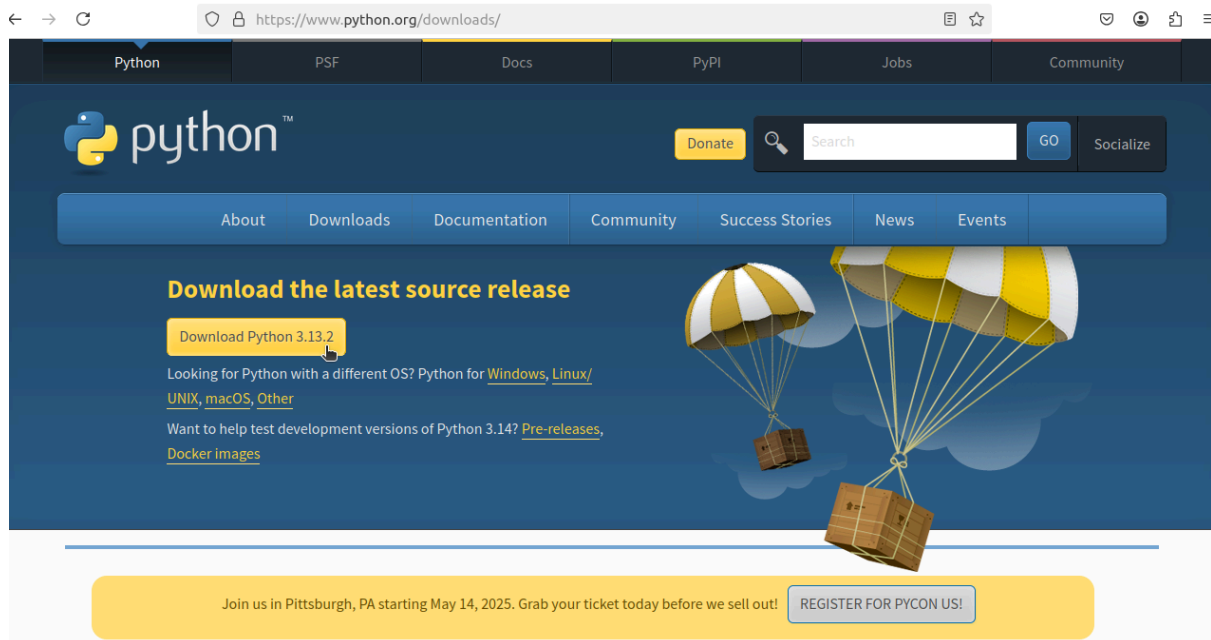
Prerequisites

Before starting, ensure you have the following:

- **Operating System:** Windows 10 or later
- **Internet Browser:** Google Chrome (recommended)
- **Stable Internet Connection**
- **Basic knowledge of Windows OS**

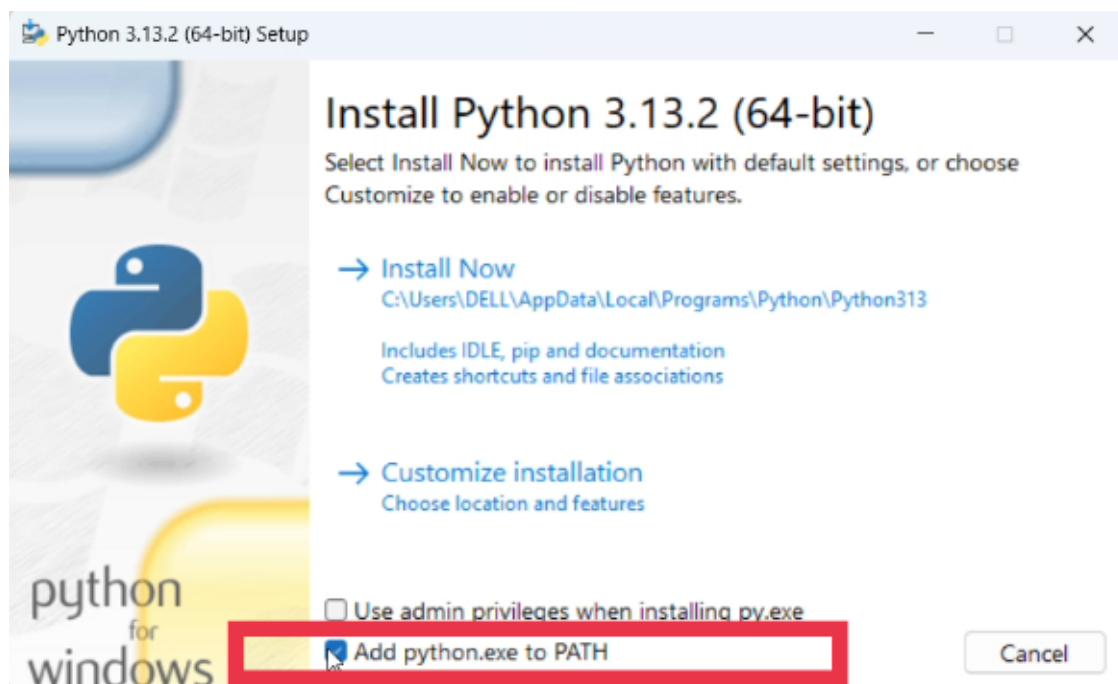
Step 1: Install Python

1. Open **Browser** and go to the [Python downloads page](https://www.python.org/downloads/).-<https://www.python.org/downloads/>
2. Click on **Download Python 3.13** (or the latest available version).

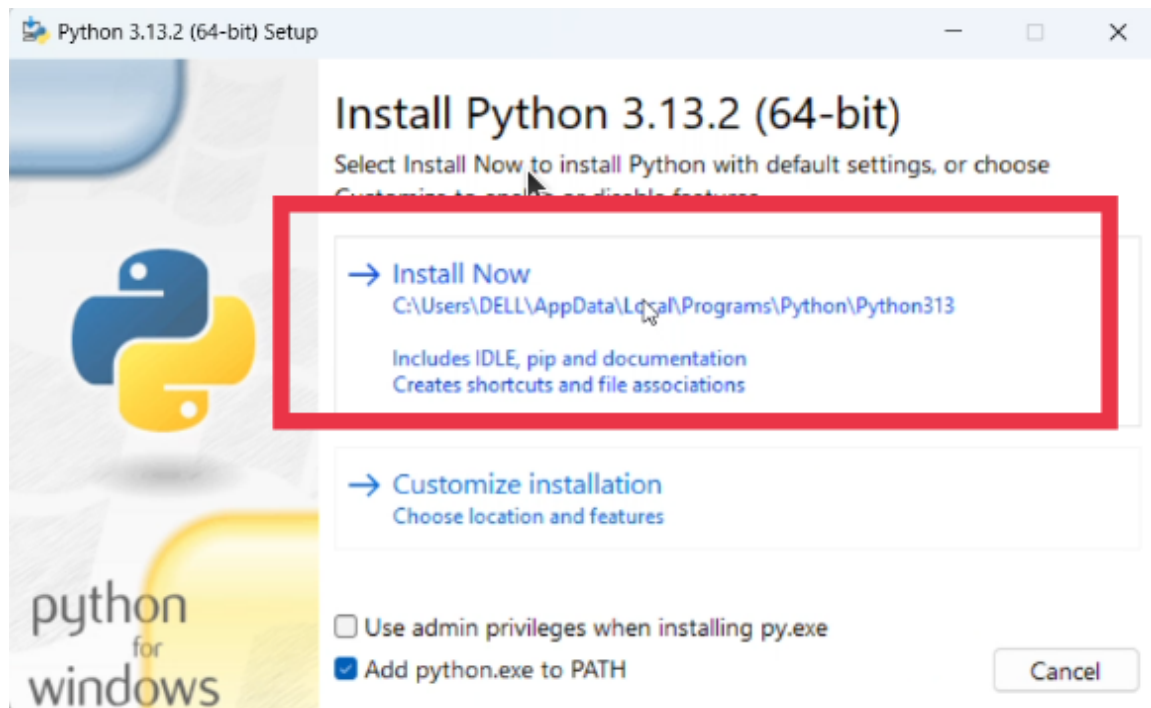


3. Choose a directory (e.g., **Downloads**) and click **OK** to start downloading.
4. Navigate to the **Downloads** folder and double-click the **Python executable file**.
5. In the installation window:

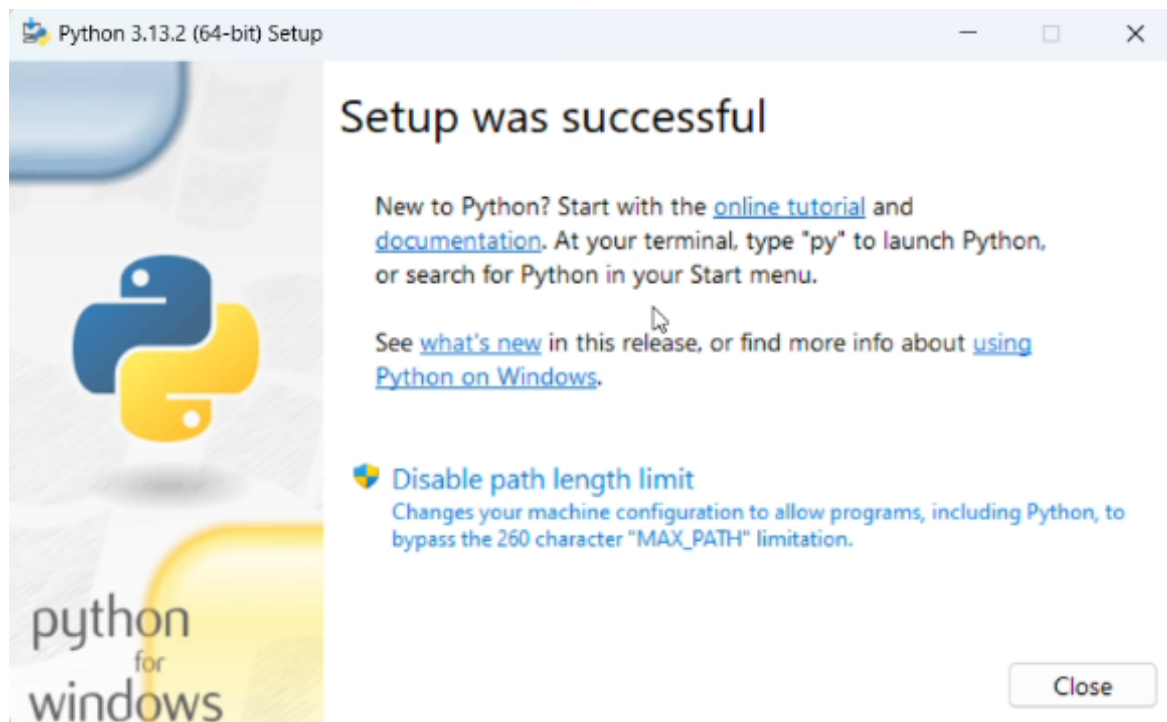
Check **"Add Python.exe to PATH"**.



Click on the **Install Now**.

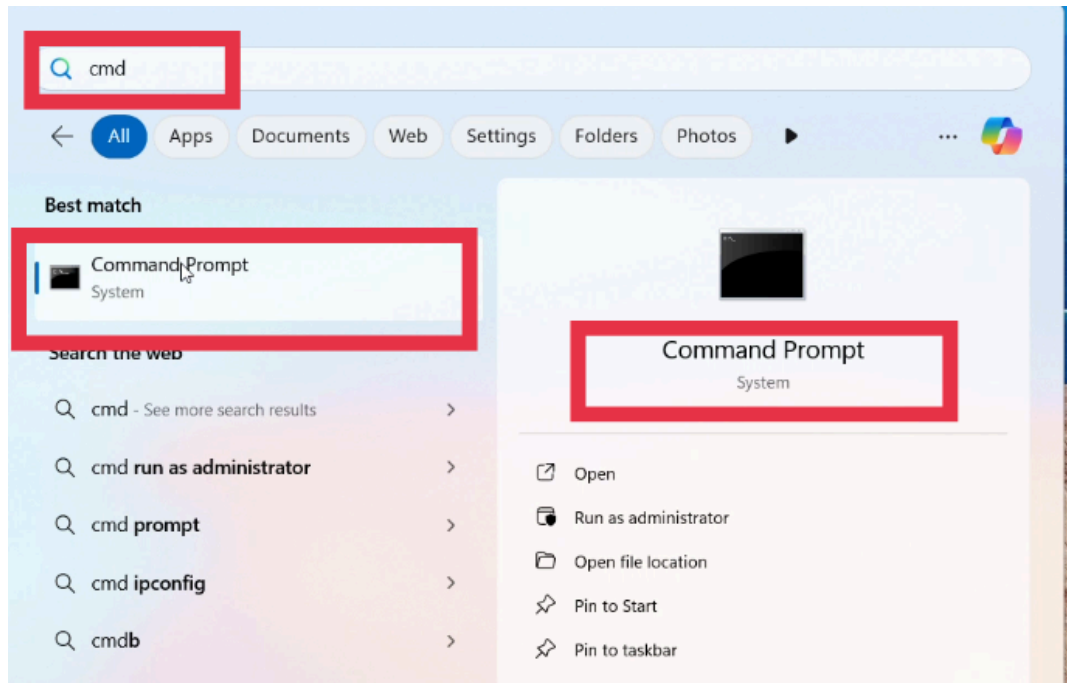


Wait for the installation to complete and look for the success message.



Verify Python Installation

1. Open **Command Prompt** (Press **Win + S**, in the search bar type **cmd**, and press **Enter**).



2. Type the following command and press Enter:
python --version

A screenshot of a Windows Command Prompt window. The title bar at the top says 'Command Prompt'. The window content shows the following text: 'Microsoft Windows [Version 10.0.22631.3880] (c) Microsoft Corporation. All rights reserved. C:\Users\DELL>python --version Python 3.13.2 C:\Users\DELL>'. The command 'python --version' and its output 'Python 3.13.2' are highlighted with a light blue selection box. A mouse cursor is visible near the end of the command line.

The installed Python version (e.g., **Python 3.13**) should be displayed as shown in above.

Step 2: Install Jupyter Notebook

1. Update `pip` (Python's package manager) by running the command prompt:

Type- `python -m pip install --upgrade pip`

```
C:\Users\DELL>python -m pip install --upgrade pip
```

2. Install Jupyter Notebook using:

`python -m pip install jupyter`

```
C:\Users\DELL>python -m pip install jupyter
```

3. Wait for the installation to finish. You should see a success message.

Step 3: Install the MicroPython Kernel

1. Open **Command Prompt** and type:

`pip3 install jupyter_micropython_kernel`

```
C:\Users\DELL>pip3 install jupyter_micropython
```

2. Register the MicroPython kernel with Jupyter by running:

`python -m jupyter_micropython_kernel.install`

```
C:\Users\DELL>python -m jupyter_micropython_kernel.install
```

3. Verify the installation by checking the available kernels:

`jupyter kernelspec list`

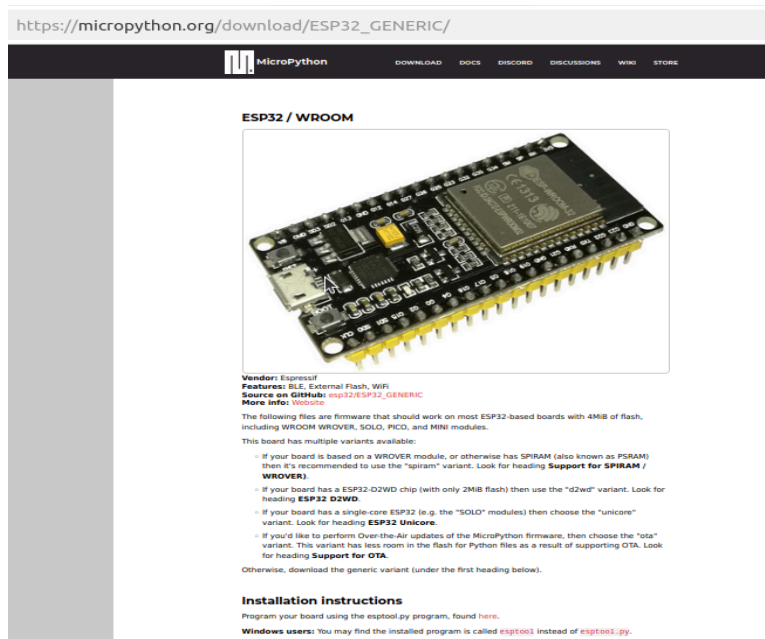
```
C:\Users\DELL>jupyter kernelspec list
Available kernels:
  micropython  C:\Users\DELL\AppData\Roaming\jupyter\kernels\micropython
  python3      C:\Users\DELL\AppData\Local\Programs\Python\Python313\share\jupyter\kernels\python3
```

You should see the MicroPython **Kernel** in the list.

Step 4: Upload MicroPython Firmware to ESP32

Download the Firmware

1. Open Google Chrome and visit the [MicroPython Firmware download page](https://micropython.org/download/ESP32_GENERIC/).-https://micropython.org/download/ESP32_GENERIC/



2. Scroll down and Click on the latest **.bin file** to download it.

Firmware

Releases

[v1.24.1 \(2024-11-29\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#) (latest)

[v1.24.0 \(2024-10-25\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.23.0 \(2024-06-02\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.22.2 \(2024-02-22\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.22.1 \(2024-01-05\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.22.0 \(2023-12-27\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.21.0 \(2023-10-05\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.20.0 \(2023-04-26\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.19.1 \(2022-06-18\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.18 \(2022-01-17\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.17 \(2021-09-02\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.16 \(2021-06-23\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.15 \(2021-04-18\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.14 \(2021-02-02\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.13 \(2020-09-02\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.12 \(2019-12-20\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

Preview builds

[v1.25.0-preview.402.g274c8c419 \(2025-03-25\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#)

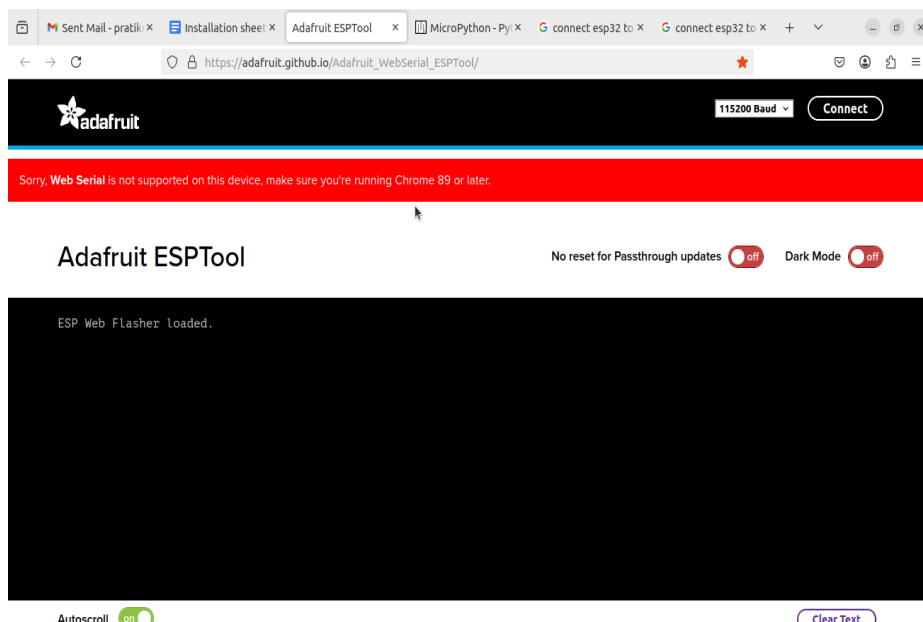
[v1.25.0-preview.393.gf1018ee5c \(2025-03-17\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#)

Connect the ESP32 to system

Connect the **ESP32 microcontroller** to your computer using a **USB cable**.



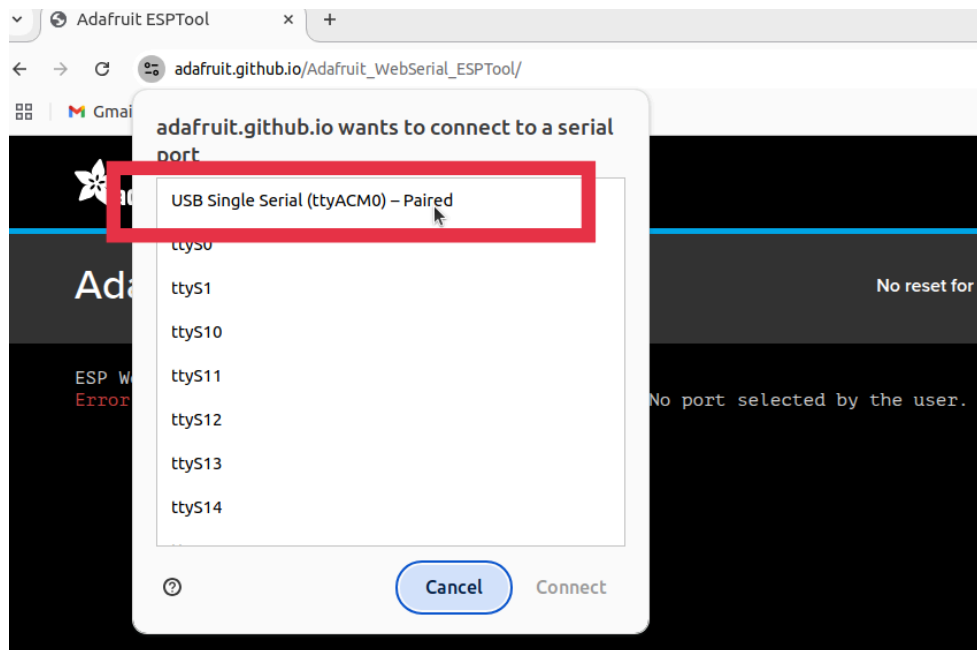
Open a new tab in Chrome and go to [Adafruit WebSerial ESPTool](https://adafruit.github.io/Adafruit_WebSerial_ESPTool/).
(https://adafruit.github.io/Adafruit_WebSerial_ESPTool/)



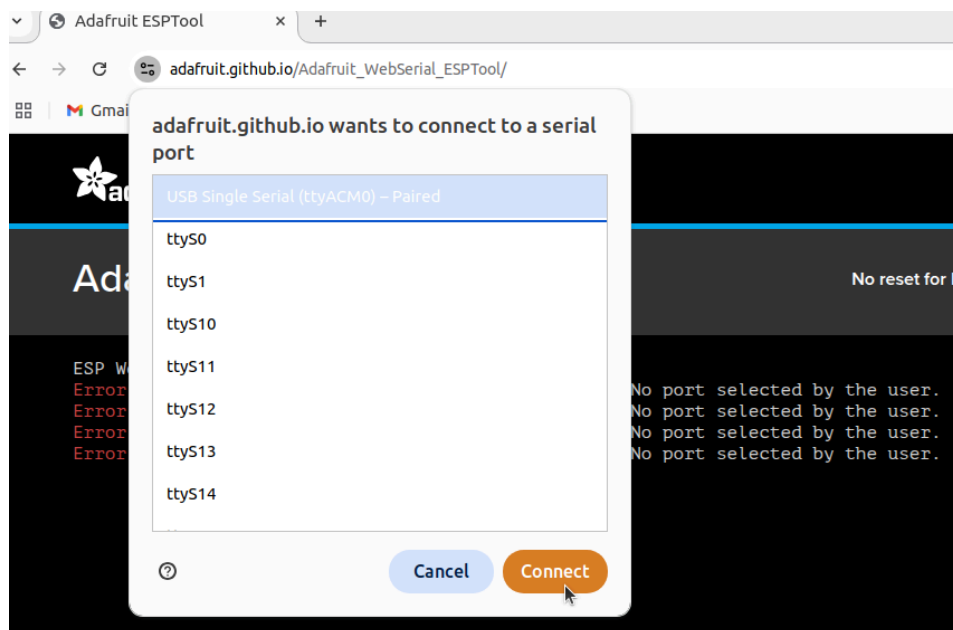
Click the **Connect** button in the top-right corner.



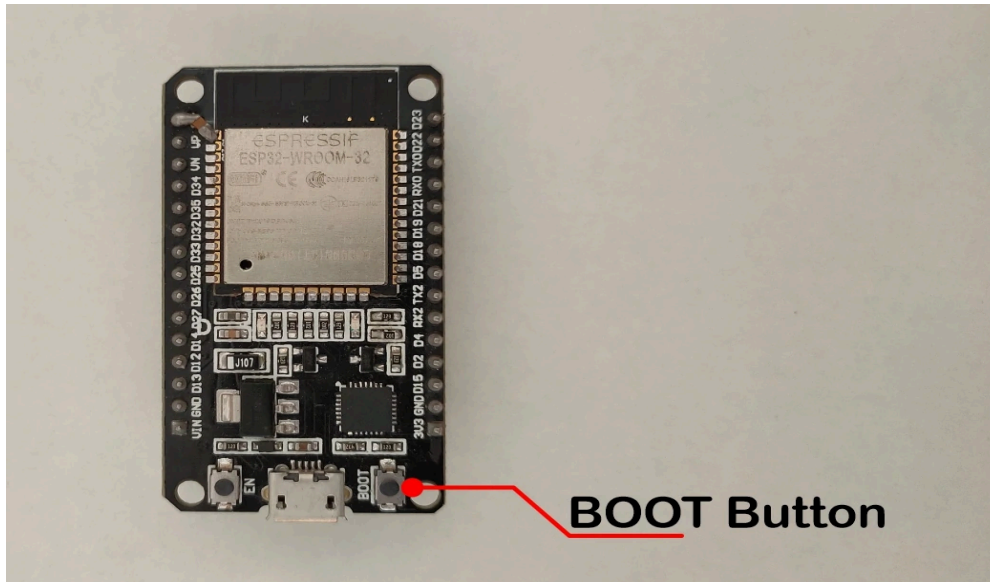
Select the USB port where ESP32 is connected.



Next click on the **Connect**.

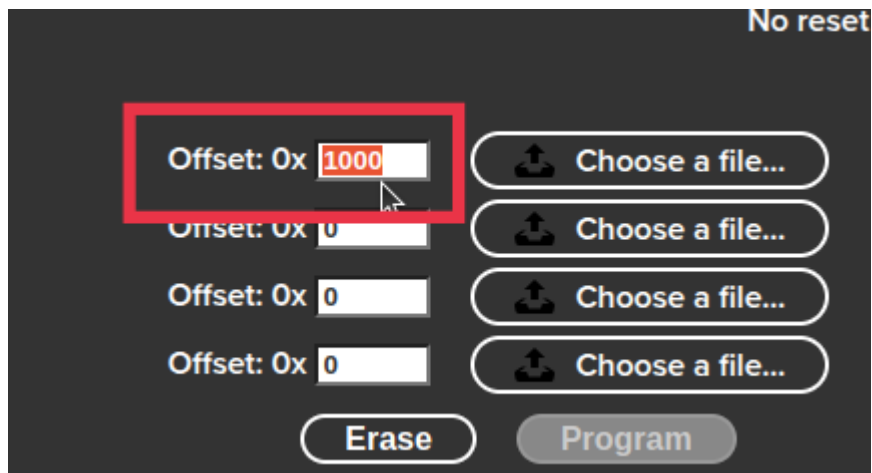


Press and hold the **BOOT** button on the ESP32 board after pressing the connect button.

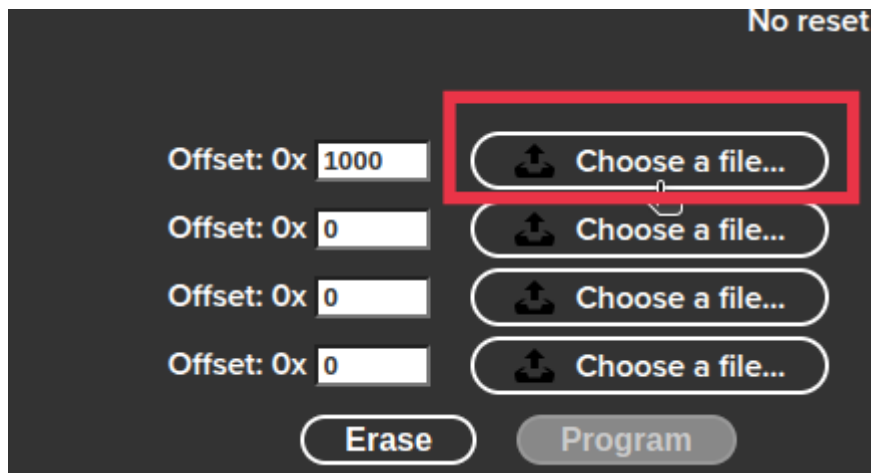


In the tool interface:

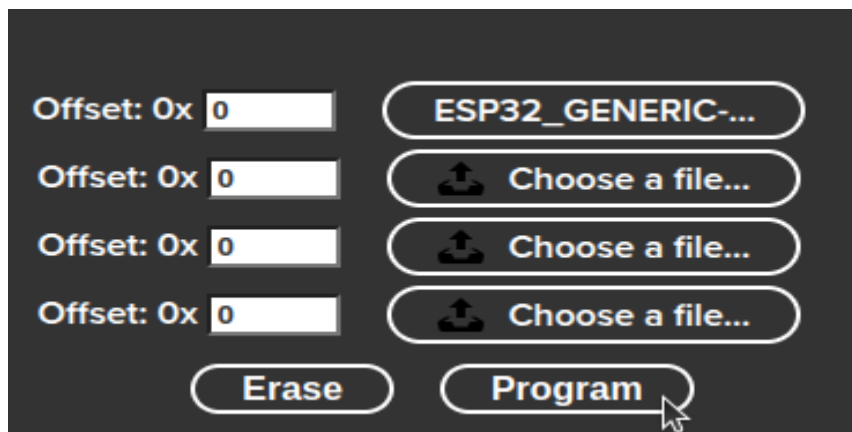
- Enter **1000** in the first offset text box.



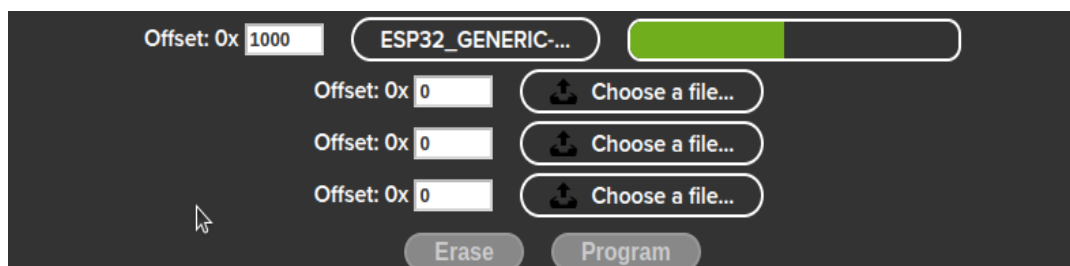
Click **Choose File** and select the **ESP32 firmware .bin file** from your download directory.



Click **Program** to upload the firmware.



Once the "Program" button is clicked, the uploading process will begin to upload the firmware to the ESP32 microcontroller, as shown below.



Wait for the upload to complete. A message will appear: "To run the new firmware, please reset your device."

```
Writing at 0x199999... (100%)
Wrote 1691664 bytes (1109554 compressed) at 0x1000 in 103.152 seconds.
Hash of data verified.
Leaving...
To run the new firmware, please reset your device.
```

This message indicates that the firmware has been successfully uploaded to the ESP32 microcontroller.

2.Installation Guide for MicroPython Kernel on Jupyter Notebook ([Linux OS](#))

Prerequisites

Before starting, ensure you have the following:

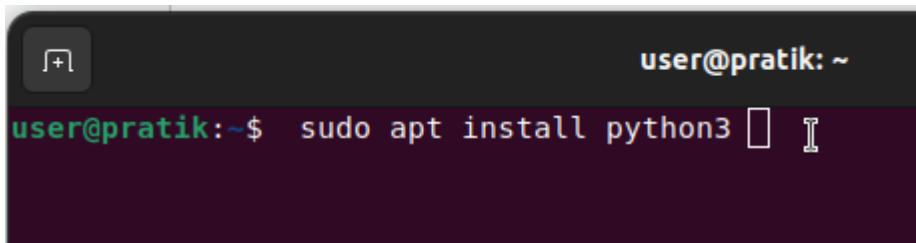
- Operating System: Ubuntu Linux OS (version 22.04 or later)
- Internet Browser: Google Chrome (recommended)
- Stable Internet Connection
- ESP32 Microcontroller
- Basic knowledge of Ubuntu Linux OS

Step 1: Install Python

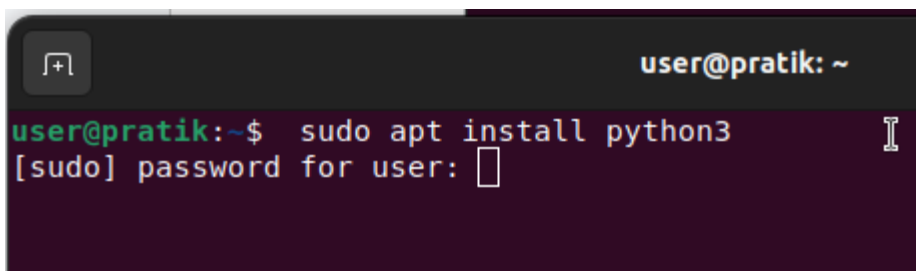
1. Open a Terminal window (Ctrl + Alt + T).

Install Python by typing the following command:

```
sudo apt install python3
```

A terminal window with a dark background. The prompt is 'user@pratik: ~'. The command 'sudo apt install python3' is typed in green text, followed by a cursor and a vertical bar.

2. Press Enter and enter your system password when prompted.

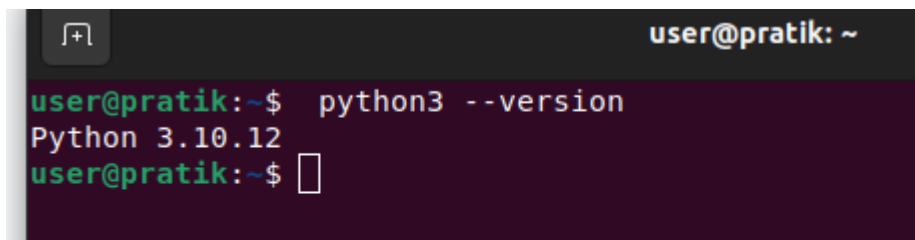
A terminal window with a dark background. The prompt is 'user@pratik: ~'. The command 'sudo apt install python3' is typed in green text. Below it, the prompt '[sudo] password for user:' is shown in red text, followed by a cursor and a vertical bar.

3. The terminal will display a list of packages to install.
4. Press Y and then Enter to confirm the installation.
5. Wait for the installation to complete.
6. Verify Python Installation

In the terminal, type the following command and press Enter:

```
python3 --version
```

The installed Python version should be displayed as shown

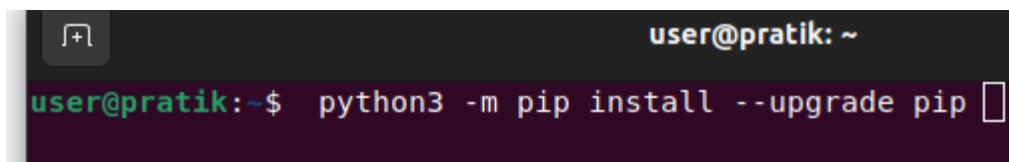
A terminal window with a dark purple background. The title bar shows a window icon and the text 'user@pratik: ~'. The prompt is 'user@pratik:~\$'. The command 'python3 --version' has been entered, and the output 'Python 3.10.12' is displayed on the next line. The prompt is followed by a white cursor box.

```
user@pratik:~$ python3 --version
Python 3.10.12
user@pratik:~$
```

Step 2: Install Jupyter Notebook

Update pip (Python's package manager) by running the command:

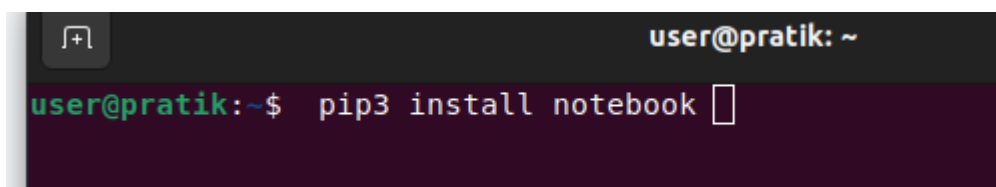
```
python3 -m pip install --upgrade pip
```

A terminal window with a dark purple background. The title bar shows a window icon and the text 'user@pratik: ~'. The prompt is 'user@pratik:~\$'. The command 'python3 -m pip install --upgrade pip' has been entered, and the prompt is followed by a white cursor box.

```
user@pratik:~$ python3 -m pip install --upgrade pip
```

Install Jupyter Notebook using:

```
pip3 install notebook
```

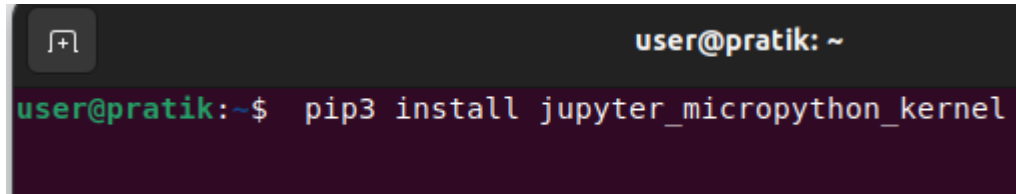
A terminal window with a dark purple background. The title bar shows a window icon and the text 'user@pratik: ~'. The prompt is 'user@pratik:~\$'. The command 'pip3 install notebook' has been entered, and the prompt is followed by a white cursor box.

```
user@pratik:~$ pip3 install notebook
```

Step 3: Install the MicroPython Kernel

Open **Terminal** and type:

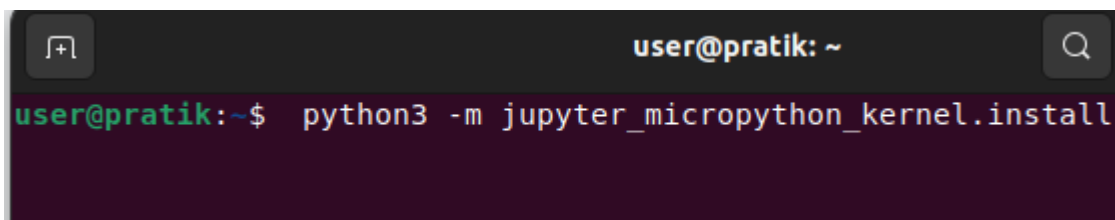
```
pip3 install jupyter_micropython_kernel
```

A terminal window with a dark background. The title bar shows 'user@pratik: ~'. The prompt is 'user@pratik:~\$'. The command 'pip3 install jupyter_micropython_kernel' is entered and partially visible.

```
user@pratik:~$ pip3 install jupyter_micropython_kernel
```

Register the MicroPython kernel with Jupyter by running:

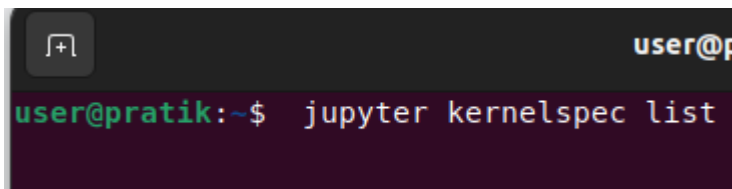
```
python3 -m jupyter_micropython_kernel.install
```

A terminal window with a dark background. The title bar shows 'user@pratik: ~' and a search icon. The prompt is 'user@pratik:~\$'. The command 'python3 -m jupyter_micropython_kernel.install' is entered.

```
user@pratik:~$ python3 -m jupyter_micropython_kernel.install
```

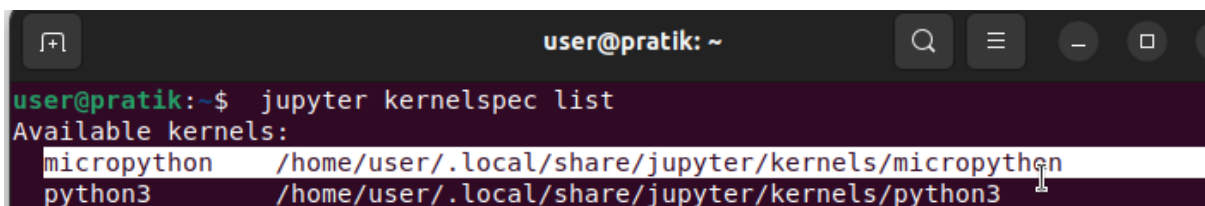
Verify the installation by checking the available kernels:

```
jupyter kernelspec list
```

A terminal window with a dark background. The title bar shows 'user@pratik: ~'. The prompt is 'user@pratik:~\$'. The command 'jupyter kernelspec list' is entered.

```
user@pratik:~$ jupyter kernelspec list
```

You should see the MicroPython Kernel in the list as shown in below,

A terminal window with a dark background. The title bar shows 'user@pratik: ~' and window control icons. The prompt is 'user@pratik:~\$'. The command 'jupyter kernelspec list' is entered, and the output is displayed. The 'micropython' entry is highlighted with a mouse cursor.

```
user@pratik:~$ jupyter kernelspec list
Available kernels:
micropython  /home/user/.local/share/jupyter/kernels/micropython
python3     /home/user/.local/share/jupyter/kernels/python3
```

Step 4: Upload MicroPython Firmware to ESP32

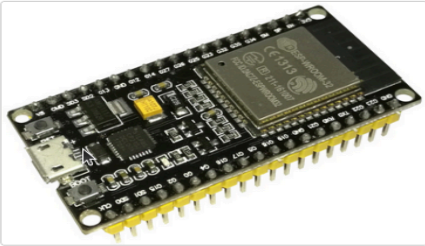
Download the Firmware

3. Open Google Chrome and visit the [MicroPython Firmware download page](https://micropython.org/download/ESP32_GENERIC/).-https://micropython.org/download/ESP32_GENERIC/

https://micropython.org/download/ESP32_GENERIC/

 **MicroPython** [DOWNLOAD](#) [DOCS](#) [DISCORD](#) [DISCUSSIONS](#) [WIKI](#) [STORE](#)

ESP32 / WROOM



Vendor: Espressif
Features: 802.11E, External Flash, WiFi
Source on GitHub: [esp32/ESP32_GENERIC](#)
More info: [WROOM](#)

The following files are firmware that should work on most ESP32-based boards with 4MiB of flash, including WROOM WROVER, SOLO, PICO, and MINI modules.

This board has multiple variants available:

- If your board is based on a WROVER module, or otherwise has SPIRAM (also known as PSRAM) then it's recommended to use the "spiram" variant. Look for heading **Support for SPIRAM / WROVER**.
- If your board has a ESP32-D2WD chip (with only 2MiB flash) then use the "d2wd" variant. Look for heading **ESP32 D2WD**.
- If your board has a single-core ESP32 (e.g. the "SOLO" modules) then choose the "unicore" variant. Look for heading **ESP32 Unicore**.
- If you'd like to perform Over-the-Air updates of the MicroPython firmware, then choose the "ota" variant. This variant has less room in the flash for Python files as a result of supporting OTA. Look for heading **Support for OTA**.

Otherwise, download the generic variant (under the first heading below).

Installation instructions

Program your board using the `esptool.py` program, found [here](#).

Windows users: You may find the installed program is called `esptool1` instead of `esptool.py`.

4. Scroll down and Click on the latest **.bin file** to download it.

Firmware

Releases

[v1.24.1 \(2024-11-29\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#) (latest)

[v1.24.0 \(2024-10-25\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.23.0 \(2024-06-02\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.22.2 \(2024-02-22\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

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[v1.20.0 \(2023-04-26\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

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[v1.13 \(2020-09-02\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

[v1.12 \(2019-12-20\) .bin](#) / [\[.elf\]](#) / [\[.map\]](#) / [\[Release notes\]](#)

Preview builds

[v1.25.0-preview.402.g274c8c419 \(2025-03-25\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#)

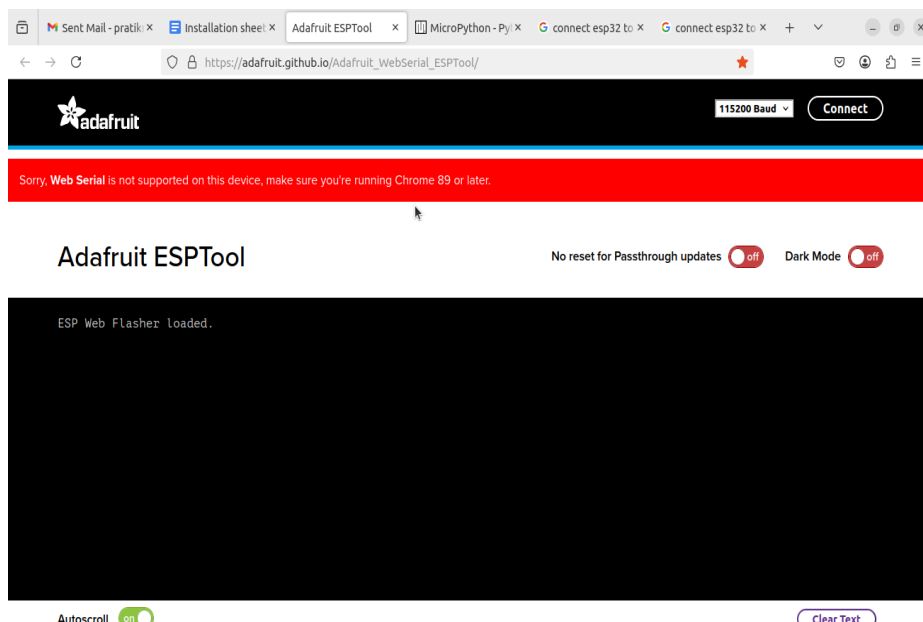
[v1.25.0-preview.393.gf1018ee5c \(2025-03-17\) .bin](#) / [\[.app-bin\]](#) / [\[.elf\]](#) / [\[.map\]](#)

Connect the ESP32 to system

Connect the **ESP32 microcontroller** to your computer using a **USB cable**.



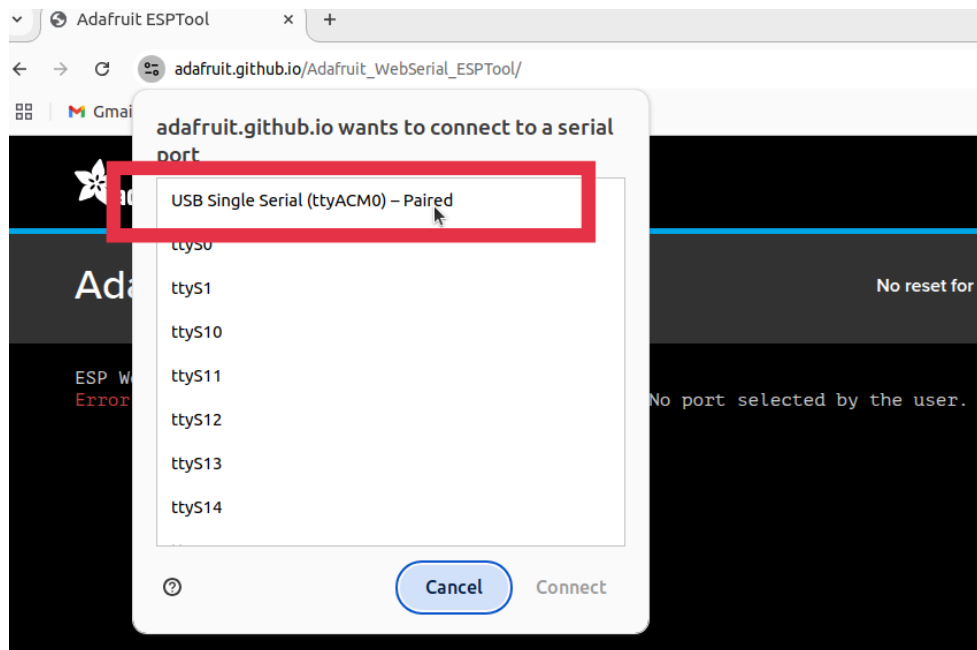
Open a new tab in Chrome and go to [Adafruit WebSerial ESPTool](https://adafruit.github.io/Adafruit_WebSerial_ESPTool/).
(https://adafruit.github.io/Adafruit_WebSerial_ESPTool/)



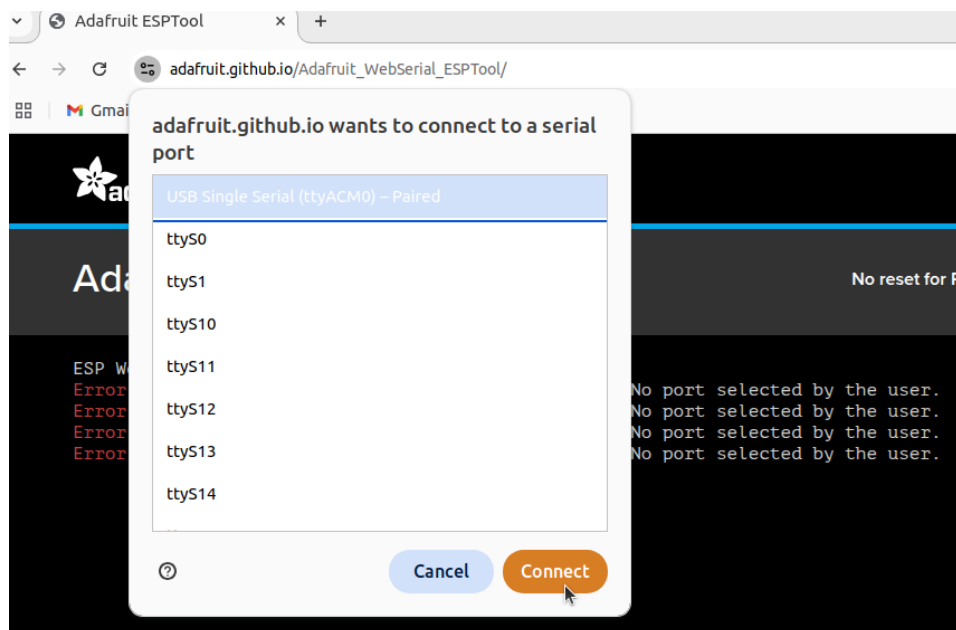
Click the **Connect** button in the top-right corner.



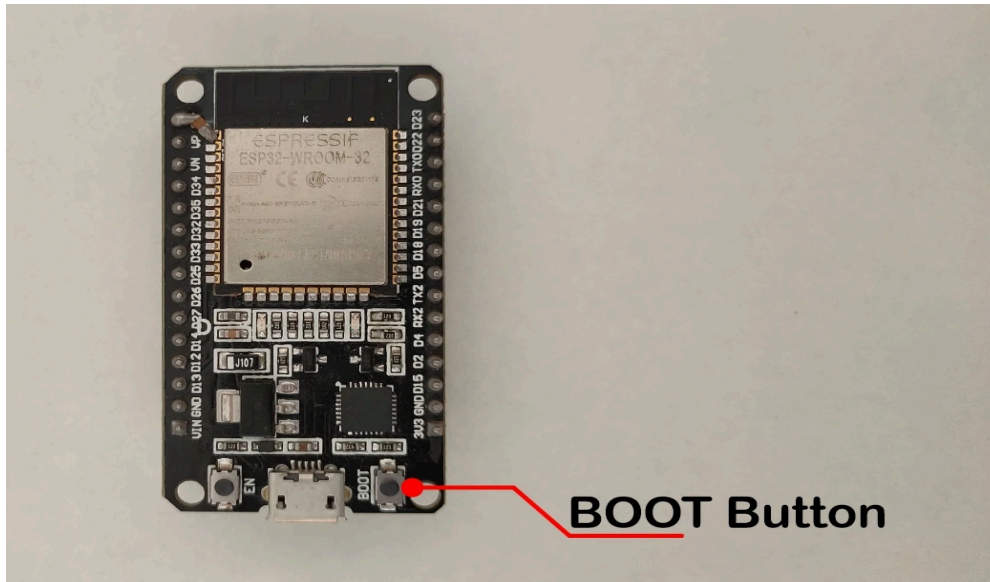
Select the USB port where ESP32 is connected.



Next click on the **Connect**.

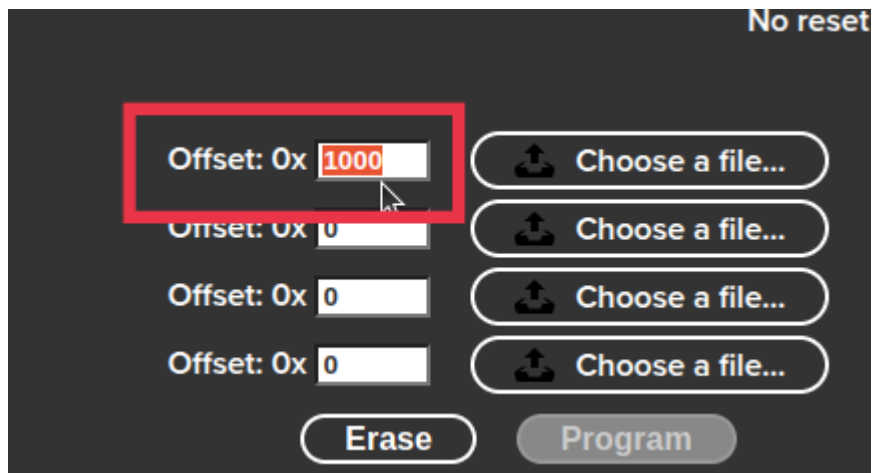


Press and hold the **BOOT** button on the ESP32 board after pressing the connect button.

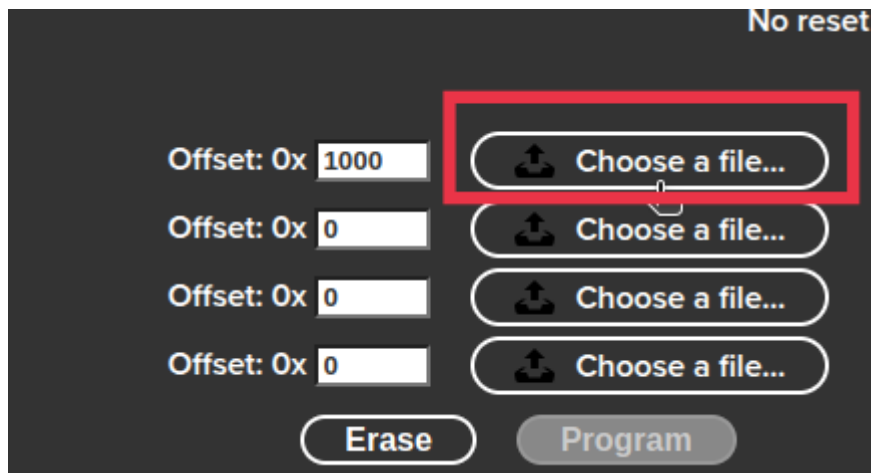


In the tool interface:

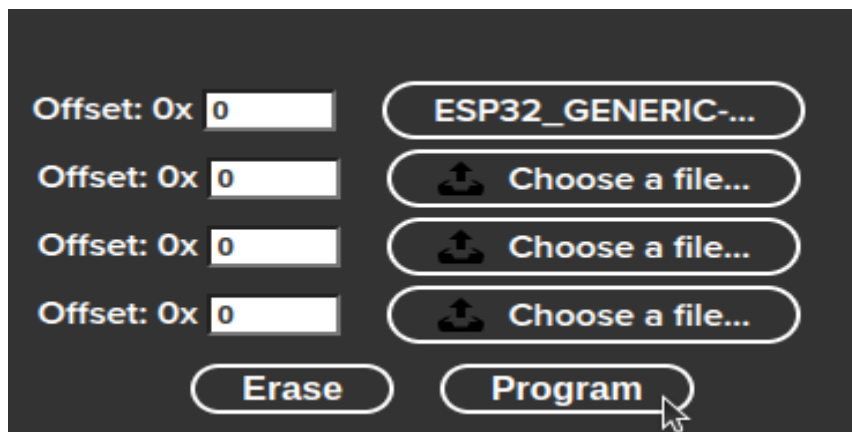
- Enter **1000** in the first offset text box.



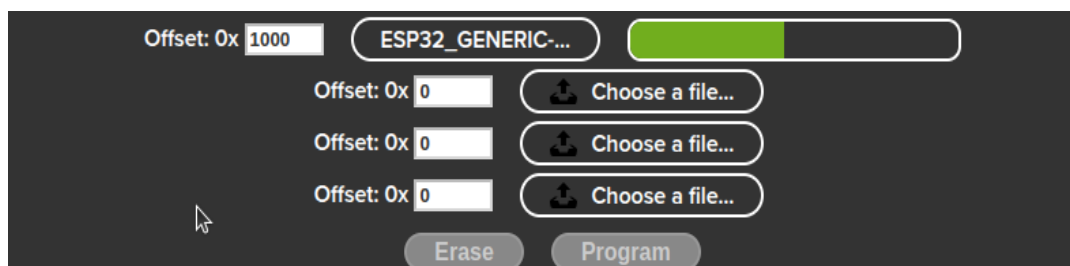
Click **Choose File** and select the **ESP32 firmware .bin file** from your download directory.



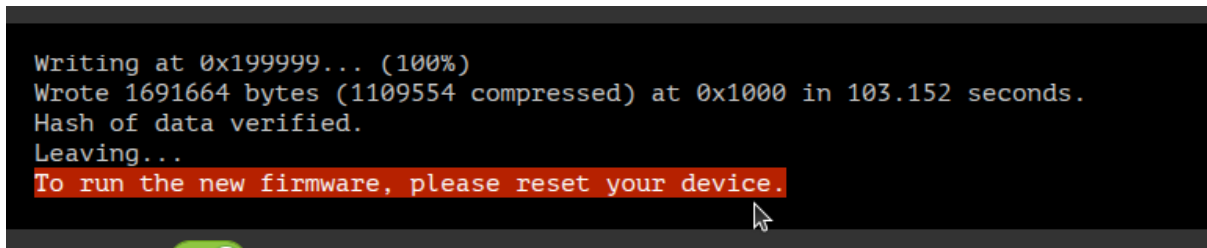
Click **Program** to upload the firmware.



Once the "Program" button is clicked, the uploading process will begin to upload the firmware to the ESP32 microcontroller, as shown below.



Wait for the upload to complete. A message will appear: "To run the new firmware, please reset your device."



```
Writing at 0x199999... (100%)
Wrote 1691664 bytes (1109554 compressed) at 0x1000 in 103.152 seconds.
Hash of data verified.
Leaving...
To run the new firmware, please reset your device.
```

This message indicates that the firmware has been successfully uploaded to the ESP32 microcontroller.

Next Steps

After successful installation, you can start using the MicroPython Experiment with Jupyter Notebook. The kernel will allow you to interact with your ESP32 directly through the notebook interface.

Troubleshooting Tips

- If Python isn't recognized in Windows, restart your computer after installation
- If ESP32 isn't recognized, check your USB connection and drivers
- Make sure to hold the BOOT button until the firmware upload begins